

A Clean Primitive-Family Atlas for the Circular Billiard

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Abstract

In the disk, the billiard map is exactly a rigid translation in the boundary angle and signed half-chord-angle coordinates. We prove one all-parameter closure theorem: every reduced fraction (m/n) , with $1 \leq m < n/2$, gives two orientation-separated clean S^1 families, with explicit chord length, caustic, action, repetitions, and unipotent return shear. The diameter and grazing faces are treated separately. The auxiliary $p = \sin \alpha$ is not called a canonical momentum. This is a geometric Route-A result; no target arithmetic or spectral match is asserted.

1 Frozen disk and incidence convention

Let $D_R = \{x^2 + y^2 < R^2\}$. At a collision write $\theta \in \mathbb{R}/2\pi\mathbb{Z}$ for boundary angle and $\alpha \in (-\pi/2, \pi/2)$ for the signed half-chord angle defined by the oriented central increment $\theta' - \theta = 2\alpha \pmod{2\pi}$. Its absolute value is the acute angle-to-tangent magnitude and its sign records direction. The auxiliary incidence amplitude $p = \sin \alpha$ records the sign and angle magnitude; it is not a canonical momentum. In the working angle chart the exact map is

$$B(\theta, \alpha) = (\theta + 2\alpha, \alpha) \pmod{2\pi}, \quad DB = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}. \quad (1)$$

The physical billiard symplectic form is expressed using the appropriate canonical boundary momentum; we make no claim that $d\theta \wedge d\alpha$ is that form.

2 Primitive rational families

Theorem 1 (complete interior rational families). *For $\gcd(m, n) = 1$ and $1 \leq m < n/2$, put $\alpha_\varepsilon = \varepsilon\pi m/n$, $\varepsilon \in \{+1, -1\}$. Then $B^n(\theta, \alpha_\varepsilon) = (\theta, \alpha_\varepsilon)$ for every θ , and n is the minimal positive bounce period. Conversely, every interior rational rotation family has this form.*

Proof. Iterating (1) gives $B^q(\theta, \alpha) = (\theta + 2q\alpha, \alpha)$. For $\alpha = \varepsilon\pi m/n$, the shift after n bounces is $\varepsilon 2\pi m$. If a smaller q closes, $qm/n \in \mathbb{Z}$, which contradicts $\gcd(m, n) = 1$. Conversely, a rational rotation in the fundamental interval has a unique reduced numerator/denominator and one of the two signs. $\square$

Proposition 2 (length, caustic, and action). *For either orientation the chord, polygon length, concentric caustic radius, and fixed-speed geometric action are*

$$\ell_{m,n} = 2R \sin(\pi m/n), \quad L_{m,n} = 2nR \sin(\pi m/n), \quad r_c = R \cos(\pi m/n), \quad S_{m,n} = p_0 L_{m,n}. \quad (2)$$

At the frozen unit speed $p_0 = 1$, $S = L$. The k -fold repetition has kn bounces and multiplies both L and S by k ; its unreduced label (km, kn) is retained but is not primitive.

Proof. The central angle between successive endpoints is $2\pi m/n$, so the chord is the corresponding isosceles-triangle base. Its distance from the center is $R \cos(\pi m/n)$, proving the caustic statement. Summing n equal chords and integrating unit-speed momentum gives (2); repetition is immediate. $\square$

3 Clean return and determinant obstruction

For a primitive row, the fixed set is the one-dimensional family $\text{Fix}(B^n) \supset \{(\theta, \alpha_\varepsilon) : \theta \in S^1\}$. In the angle chart,

$$DB^n = \begin{pmatrix} 1 & 2n \\ 0 & 1 \end{pmatrix}, \quad DB^n - I = \begin{pmatrix} 0 & 2n \\ 0 & 0 \end{pmatrix}. \quad (3)$$

Hence

$$\ker(DB^n - I) = \text{span}\{(1, 0)\} = T(S_\theta^1), \quad \det(I - DB^n) = 0. \quad (4)$$

The return is unipotent, with both eigenvalues one. Thus an isolated-orbit determinant denominator is obstructed: the clean family must not be assigned an isolated amplitude. The finite receipt verifies the kernel itself, not only the vanishing determinant.

Table 1: Selected rows from the finite exact/high-precision receipt (R=1).

(m, n)	orientation	α	ℓ	$L = S$	r_c
(1, 3)	+	$\pi/3$	1.7321	5.1962	0.5000
(1, 4)	−	$-\pi/4$	1.4142	5.6569	0.7071
(2, 5)	+	$2\pi/5$	1.9021	19.0211	0.3090
(3, 7)	−	$-3\pi/7$	1.9499	27.2986	0.2225

4 Revision-one ledger

The producer enumerates every primitive pair through $n = 12$, both signs, six explicit repetitions, and the endpoint rows. Chebyshev relations $T_n(\cos(\pi m/n)) = (-1)^m$ provide an algebraic/high-precision receipt. An independent checker makes more than two thousand assertions and a SymPy script repeats the trigonometric and shear identities.

5 Diameter, grazing, and natural quantization

The diameter case $(m, n) = (1, 2)$, $\alpha = \pm\pi/2$, is one endpoint-equivalent family, not two families. Its angle-chart return matrix for two bounces is the boundary value $\begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$, but it is not an interior regular row. At $\alpha = 0$, $p = 0$, the chord has zero length; the receipt records one grazing row with two one-sided oriented limits. No boundary degeneration is silently counted as a new primitive flight.

For A4 we record only the natural finite quantizations of the same disk: $-\Delta_{D_R}$ with $f|_{\partial D_R} = 0$ (Dirichlet), or with $\partial_\nu f|_{\partial D_R} = 0$ (Neumann). This is a definition of a source-local quantization problem, not an eigenvalue table or a target match.

References

- [1] G. D. Birkhoff, “On the periodic motions of dynamical systems,” *Acta Mathematica* 50 (1927), 359–379, DOI 10.1007/BF02421325.
- [2] R. L. Bishop, “Circular Billiard Tables, Conjugate Loci, and a Cardioid,” *Regular and Chaotic Dynamics* 8 (2003), 83–95, DOI 10.1070/RD2003v008n01ABEH000227.